1. Sampling & Estimation

1.1. Sample & Population

- **Population:** collection of all items
- **Sample:** subset of population used as a representation of the entire population

1.2. Central Limit Theorem

- If $(X_1, X_2, \ldots, X_n)$ is a random sample of size n drawn from any population with mean μ and variance σ^2 then the sample has:

Expected mean, μ

Expected variance, $\frac{\sigma^2}{n}$

It forms a normal distribution:

$$\widetilde{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

{IS Ex 10f} Question 12:

The weights of the trout at a trout farm are normally distributed with mean 1kg & standard deviation 0.25kg

1. Find, to 4 decimal places, the probability that a trout chosen at random weighs more than 1.25kg.
2. If $\overline{Y}$kg represents mean weight of a sample of 10 trout chosen at random, state the distribution of $\overline{Y}$: evaluate the mean and variance. Find the probability that the mean weight of a sample of 10 trout will be less than 0.9kg

Solution:

Part (a):

Write down distribution

$$X \sim N\left(1,\ 0.25^2\right)$$

Write down the probability they want

$$P(X > 1.25) = 1 - P(X < 1.25)$$

Standardize and evaluate

$$1 - P\left(Z < \frac{1.25 - 1}{0.25}\right) = 0.1587$$

Part (b):

Write down initial distribution

$$X \sim N\left(1,\ 0.25^2\right)$$

For sample, mean remains equal but variance changes

Find new variance

Variance of sample $= \frac{\sigma^2}{n} = \frac{0.25^2}{10} = 0.00625$

Write down distribution of sample

$$\overline{Y} \sim N(1,\ 0.00625)$$

Write down the probability they want

$$P\left(\overline{Y} < 0.9\right)$$

Standardize and evaluate

Standardized probability is negative so do 1 minus

$$P\left(Z < \frac{0.9 - 1}{\sqrt{0.00625}}\right) = 1 - P\left(Z < \frac{0.1}{\sqrt{0.00625}}\right) = 0.103$$

1.3. Point Estimate & Confidence Interval

- A **point estimate** is a numerical value calculated from a set of data (sample) which is used as an estimate of an unknown parameter in a population
- Examples of point estimates are:

Sample mean $\overline{x}$	population mean μ
Sample proportion $\frac{r}{n}$	population proportion p
Sample variance s^2	population variance σ^2

- Point estimate close to population value but not exact
- We can determine a **confidence interval** where the population value is likely to lie in $(\overline{x} - \delta,\ \overline{x} + \delta)$

1.4. The Variance

- Variance can be calculated/given for either a sample or a population and there is a difference between them

Using the divisor $\mathbf{n}$

- This is appropriate to use when
  - data is given for the whole population and you are interested in the variance of the whole
  - data is given for the sample and you are interested in the variance of just the sample

$$\sigma^2 = \frac{1}{n}\left(\Sigma x^2 - \frac{(\Sigma x)^2}{n}\right)$$

Using the divisor $(\mathbf{n} - \mathbf{1})$

- Appropriate to use when data is given for a sample and you are estimating variance of the whole population
- The quantity calculated s^2 is known as the **unbiased estimate of the population variance**

$$s^2 = \frac{1}{n-1}\left(\Sigma x^2 - \frac{(\Sigma x)^2}{n}\right)$$

1.5. Percentage Points for a Normal Distribution

- The percentage points are determined by finding the z-value of specific percentages

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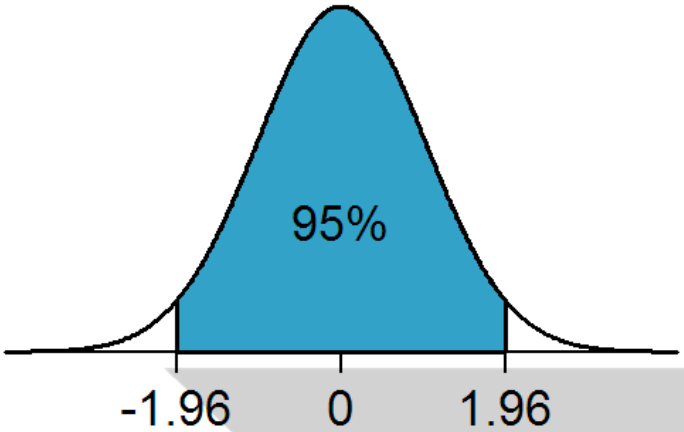
- E.g. to find the z-value of a 95% confidence level, we can see that the 5% would be removed equally from both sides (2.5%) so the z-value we would actually be finding would be of $100\% - 2.5\% = 97.5\%$



Percentage Points Table

Confidence level	90%	95%	98%	99%
z-value	1.645	1.960	2.326	2.576

1.6. Confidence Interval for a Population Mean

Sample taken from a normal population distribution with known population variance

$$\left(\overline{x} - z\frac{\sigma}{\sqrt{n}}, \overline{x} + z\frac{\sigma}{\sqrt{n}}\right)$$

- z is the value corresponding to the confidence level required and n is the sample size
- The confidence interval calculated is exact

Large sample taken from an unknown population distribution with known population variance

- By the Central Limit Theorem, the distribution of $\overline{X}$ will be approximately normal so same method as above

$$\left(\overline{x} - z\frac{\sigma}{\sqrt{n}}, \overline{x} + z\frac{\sigma}{\sqrt{n}}\right)$$

- The confidence interval calculated is an approximate

Large sample taken from an unknown population distribution with unknown population variance

- As the population variance is unknown, you must first estimate the population variance, s, using sample data

$$\left(\overline{x} - z\frac{s}{\sqrt{n}}, \overline{x} + z\frac{s}{\sqrt{n}}\right)$$

- The confidence interval calculated is an approximate

{W13-P71} Question 2:

Heights of a certain species of animal are normally distributed with $\sigma =$ 0.17m. Obtain a 99% confidence interval for the population mean, with total width less than 0.2m. Find the smallest sample size required.

Solution:

For a 99% confidence interval, find z where

$\Phi(z) =$ *0.995 (think of the 1% cut from both sides)*

$$z = 2.576$$

Subtract the limits of the interval and equate to 0.2

$$\left(\overline{x} + z\frac{\sigma}{\sqrt{n}}\right) - \left(\overline{x} - z\frac{\sigma}{\sqrt{n}}\right) = 0.2$$

$$2\left(z\frac{\sigma}{\sqrt{n}}\right) = 0.2$$

Substitute information given and find n

$$\sqrt{n} = \frac{2 \times 2.576}{0.2} \times 0.17$$

$$n = 19.2 \approx 20$$

1.7. Confidence Interval for a Population Proportion

- Calculating the confidence interval from a random sample of n observations from a population in which the proportion of successes is p and the proportion of failures is q
- The observed proportion of success $\hat{p}$ is $\frac{r}{n}$ where r represents the number of successes

$$\left(\hat{p} - z\sqrt{\frac{\hat{p}\hat{q}}{n}}, \hat{p} + z\sqrt{\frac{\hat{p}\hat{q}}{n}}\right)$$

{S10-P71} Question:

A random sample of n people were questioned about their internet use. 87 of them had a high-speed internet connection. A confidence interval for the population proportion having a high-speed internet connection is 0.1129 $< p <$ 0.1771.

1. Write down the mid-point of this confidence interval and hence find the value of n.
2. This interval is an α% confidence interval. Find α.

Solution:

Part (i):

Find the midpoint of the limits, finding p

$$0.1129 + \frac{0.1771 - 0.1129}{2} = 0.145$$

The midpoint is equal to the proportion of people with high-speed internet use so

$\frac{87}{n} = 0.145 \quad \therefore n = 600$

Part (ii):

Using the upper limit, this was calculate by:

$$0.1771 = 0.145 + z\sqrt{\frac{\text{pq}}{n}}$$

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Substituting values calculated ($q = 1 - p$), find z

$$0.0321 = z\sqrt{\frac{\frac{87}{600} \times \frac{513}{600}}{600}} \qquad \therefore z = 2.233$$

Use normal tables and find corresponding probability

$$\Phi(z) = 0.9872$$

Think of symmetry, the same area is chopped off from both sides of the graph so

$$1 - 2(1 - 09872) = 0.9744$$

Hence the α% confidence is $= 97.44\%$

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